

1 Robust conditional probability estimation

Langford’s COLT 2010 open problem [2] asks for a reduction from *multiclass* conditional probability estimation to *binary* regression, Langford asks whether there is a reduction that simultaneously obtains constant regret and has only $O(\log(k))$ runtime/oracle calls.

1.1 Setting and main result

Conditional probability estimation. A CPE problem is given by a distribution D over $\mathcal{X} \times [k]$ with conditional law $D(y | x) = \Pr_D[Y = y | X = x]$. We write $P(y | x) := D(y | x)$ throughout. A CPE *solver* is a map $A : (x, y) \mapsto \hat{p}_y(x) \in [0, 1]$ estimating $P(y | x)$; its squared-loss CPE regret is the per-label squared error against the truth,

$$\text{reg}(D, A) := \mathbb{E}_{(x,y) \sim D} \left[(A(x, y) - P(y | x))^2 \right], \quad (1.1)$$

Note the expectation is over $(x, y) \sim D$, so the label y is drawn from the same marginal that defines the target $P(\cdot | x)$.

Binary regression. A binary-regression problem is a distribution D' over $\mathcal{Z} \times [0, 1]$, where $\mathcal{Z}$ is a feature space and labels lie in $[0, 1]$ (real-valued, or Bernoulli—squared-loss regret is identical for the two, see below). A binary regressor is a map $B : \mathcal{Z} \rightarrow [0, 1]$, with Bayes-optimal predictor $\mu(z) = \mathbb{E}_{D'}[\text{label} | z]$ and squared-loss regret

$$\begin{aligned} \text{reg}(D', B) &:= \mathbb{E}_{z \sim D'} [(B(z) - \mu(z))^2] \\ &= \mathbb{E}_{(z,t) \sim D'} [(B(z) - t)^2] - \mathbb{E}_{(z,t) \sim D'} [(\mu(z) - t)^2]. \end{aligned} \quad (1.2)$$

Because $\mathbb{E}_{(z,t)}[(B(z) - t)^2] = \mathbb{E}_z[(B(z) - \mu(z))^2] + \mathbb{E}_{(z,t)}[(\mu(z) - t)^2]$ by the bias-variance split.

A reduction. A reduction is a pair (R, R^{-1}) . The *training map* R turns each CPE example $(x, y) \sim D$ into one or more binary-regression examples, defining the induced distribution D' ; a regressor B is trained on D' . The *decoding map* R^{-1} answers a CPE query (x, y) by issuing oracle calls to B and combining the answers into $\hat{p}_y(x) \in [0, 1]$.

The reduction’s quality is measured by the regret-transfer bound C

$$\text{reg}(D, R^{-1}) \leq C \text{reg}(D', B)$$

which must hold for every distribution D and every oracle algorithm B for binary regression; The cost of the reduction is the runtime of R and R^{-1} .

Remark 1.1 (Runtime vs Oracle calls). It is unclear whether runtime in [2] refers to the number of oracle calls or the actual computation time in word RAM model. In our proof, we explicitly count the oracle calls and computation time separately. We lean towards runtime $\approx$ oracle calls since the ECOC/PECOC approaches need $\Omega(k)$ oracle calls and each oracle contains an input of at least $\Omega(k)$ bits, but in [2], it counts as $\Theta(k)$ runtime.

As pointed out in [2], the two known endpoints miss on opposite axes: the binary tree approach has $O(\log k)$ runtime but loses a $O((\log k)^2)$ factor in regret, while the error correction code approach ECOC/PECOC [3] achieves a constant factor in regret but costs $\Omega(k)$ runtime.

Langford asks whether there is a reduction that has only constant regret blow up and $O(\log k)$ runtime.

This section gives a resolution to the open question. We exhibit a randomized reduction (using matching-vector code) that achieves the constant-regret (with an explicit universal constant $C = 8$), using $O(\log k)$ oracle calls and $O((\log k)^2)$ computation time.

Reduction	Regret transfer	Oracle call
Regression tree	$O((\log k)^2)$	$\Theta(\log k)$
ECOC/PECOC [3]	$O(1)$	$\Theta(k)$
This section (Theorem 1.2)	8	$\Theta(\log k)$

Theorem 1.2 (Constant-regret transfer at polylog time). *For every number of classes $k \geq 2$ there is a randomized reduction pair (R, R^{-1}) from k -class conditional probability estimation to binary regression with targets in $[0, 1]$ such that, for every data distribution D over $\mathcal{X} \times [k]$ and every binary regressor B ,*

$$\text{reg}(D, R^{-1}) \leq 8 \text{reg}(D', B), \quad (1.3)$$

where D' is the induced binary-regression distribution. The decoder R^{-1} uses $\Theta(\log k)$ oracle calls per query. The running time of R^{-1} is $\Theta(\log^2 k)$.

1.2 The matching-vector code

The reduction utilizes the matching-vector code [1]. Fix the smallest even integer d with $\binom{d}{d/2} \geq k$; since $\binom{d}{d/2} = \Theta(2^d / \sqrt{d})$ this gives $d = O(\log k)$. Assign to each label $y \in [k]$ a distinct set $A_y \subseteq [d]$ of size $|A_y| = d/2$, and set

$$u_y := \mathbf{1}_{A_y} \in \{0, 1\}^d, \quad v_y := \mathbf{1}_{[d] \setminus A_y} \in \{0, 1\}^d. \quad (1.4)$$

Choose the modulus $m \geq d + 1$ to be a prime number and let $\omega := e^{2\pi i/m}$, a primitive m -th root of unity. All inner products $\langle u_j, w \rangle$ below are read modulo m in the exponent of ω . For a query point $w \in \mathbb{Z}_m^d$ define the complex code symbol

$$F_x(w) := \sum_{j=1}^k P(j | x) \omega^{\langle u_j, w \rangle} \in \mathbb{C}, \quad |F_x(w)| \leq \sum_j P(j | x) = 1. \quad (1.5)$$

The decoder will recover a single coefficient $P(y | x)$ of this character sum by integrating F_x along a structured line. The line direction is v_y , and the following *matching-vector property* is what makes the integration single out the $j = y$ term.

Lemma 1.3 (Matching-vector property). *With u_y, v_y as in (1.4), one has $\langle u_y, v_y \rangle = 0$, while for every $j \neq y$,*

$$\langle u_j, v_y \rangle = |A_j \setminus A_y| \in \{1, \dots, d/2\} \quad \text{is} \quad \not\equiv 0 \pmod{m}.$$

Proof. Since $u_y = \mathbf{1}_{A_y}$ and $v_y = \mathbf{1}_{[d] \setminus A_y}$ have disjoint supports, $\langle u_y, v_y \rangle = |A_y \cap ([d] \setminus A_y)| = 0$. For $j \neq y$, $\langle u_j, v_y \rangle = |A_j \cap ([d] \setminus A_y)| = |A_j \setminus A_y|$. As $|A_j| = d/2$, this is at most $d/2$. It is at least 1: if $A_j \setminus A_y = \emptyset$ then $A_j \subseteq A_y$, and since $|A_j| = |A_y| = d/2$ this forces $A_j = A_y$, contradicting $j \neq y$ (the A 's are distinct). Thus $\langle u_j, v_y \rangle \in \{1, \dots, d/2\}$, and $0 < \langle u_j, v_y \rangle \leq d/2 < d + 1 \leq m$, so it is nonzero modulo m . $\square$

The single nontrivial property of ω we use is the geometric (character) sum identity: for an integer c ,

$$\frac{1}{m} \sum_{\lambda=0}^{m-1} \omega^{\lambda c} = \begin{cases} 1, & c \equiv 0 \pmod{m}, \\ 0, & c \not\equiv 0 \pmod{m}, \end{cases} \quad (1.6)$$

1.3 The reduction (R, R^{-1})

Training map R . The binary feature space is $\mathcal{Z} = \mathcal{X} \times \mathbb{Z}_m^d \times \{\text{Re}, \text{Im}\}$, i.e. a context x , a code point w , and a part-selector. Given a CPE example $(x, y) \sim D$, R draws $w \sim \text{Unif}(\mathbb{Z}_m^d)$ and a fair coin $s \in \{\text{Re}, \text{Im}\}$, and emits the single binary example

$$\left((x, w, s), t_s \right), \quad t_{\text{Re}} = \frac{1}{2}(1 + \text{Re } \omega^{\langle u_y, w \rangle}), \quad t_{\text{Im}} = \frac{1}{2}(1 + \text{Im } \omega^{\langle u_y, w \rangle}). \quad (1.7)$$

Both targets lie in $[0, 1]$ because $|\omega^{\langle u_y, w \rangle}| = 1$ forces $\text{Re}, \text{Im} \in [-1, 1]$. This defines D' as the law of $((x, w, s), t_s)$.

Lemma 1.4 (Induced conditional means). *For the distribution D' of (1.7), the Bayes-optimal regressor satisfies*

$$\mu(x, w, \text{Re}) = \frac{1}{2}(1 + \text{Re } F_x(w)), \quad \mu(x, w, \text{Im}) = \frac{1}{2}(1 + \text{Im } F_x(w)).$$

Consequently, defining from any regressor B the complex estimate

$$\hat{F}_x(w) := (2B(x, w, \text{Re}) - 1) + i(2B(x, w, \text{Im}) - 1), \quad (1.8)$$

the Bayes regressor reconstructs $F_x(w)$ exactly: $\hat{F}_x(w) = F_x(w)$ when $B = \mu$.

Proof. Condition on (x, w, Re) . The label is $t_{\text{Re}} = \frac{1}{2}(1 + \text{Re } \omega^{\langle u_Y, w \rangle})$ with $Y \sim P(\cdot \mid x)$. By linearity of expectation and (1.5), $\mu(x, w, \text{Re}) = \frac{1}{2}(1 + \mathbb{E}_Y[\text{Re } \omega^{\langle u_Y, w \rangle}]) = \frac{1}{2}(1 + \text{Re } F_x(w))$, and identically for Im . Substituting $B = \mu$ into (1.8) gives $\hat{F}_x(w) = \text{Re } F_x(w) + i \text{Im } F_x(w) = F_x(w)$. $\square$

Decoding map R^{-1} . To answer a CPE query (x, y) , draw $W \sim \text{Unif}(\mathbb{Z}_m^d)$, query the m line points $W + \lambda v_y$ for $\lambda = 0, \dots, m-1$ (each in both parts, via $\hat{F}_x$ of (1.8)), and output

$$\hat{p}_y(x) = \text{Re} \left(\omega^{-\langle u_y, W \rangle} \cdot \frac{1}{m} \sum_{\lambda=0}^{m-1} \hat{F}_x(W + \lambda v_y) \right), \quad (1.9)$$

This uses $2m = \Theta(\log k)$ oracle calls.

1.4 Proof of Theorem 1.2

We first prove a special case: when the binary regression solver have zero regret (i.e., $B = \mu$), the the estimator (1.9) has zero error.

Lemma 1.5 (Exact decoding). *If $B = \mu$ (zero binary regret), then for every x , every label y , and every realization of $W \in \mathbb{Z}_m^d$, the un-clipped quantity inside (1.9) equals $P(y \mid x)$ exactly. Hence $\hat{p}_y(x) = P(y \mid x)$ and $\text{reg}(D, R^{-1}) = 0$, with no sampling variance.*

Proof. By Lemma 1.4, $\widehat{F}_x = F_x$. Expanding (1.5) along the line and exchanging the finite sums,

$$\frac{1}{m} \sum_{\lambda=0}^{m-1} F_x(W + \lambda v_y) = \sum_{j=1}^k P(j \mid x) \omega^{\langle u_j, W \rangle} \underbrace{\frac{1}{m} \sum_{\lambda=0}^{m-1} \omega^{\lambda \langle u_j, v_y \rangle}}_{=: S_j}.$$

By Lemma 1.3, $\langle u_y, v_y \rangle = 0$, so the geometric-sum identity (1.6) gives $S_y = 1$; while for $j \neq y$, $\langle u_j, v_y \rangle \not\equiv 0 \pmod{m}$, so $S_j = 0$. Only the $j = y$ term survives:

$$\frac{1}{m} \sum_{\lambda=0}^{m-1} F_x(W + \lambda v_y) = P(y \mid x) \omega^{\langle u_y, W \rangle}.$$

Multiplying by $\omega^{-\langle u_y, W \rangle}$ yields the real number $P(y \mid x)$, whose real part is $P(y \mid x) \in [0, 1]$; clipping is inert. Averaging the zero error over $(x, y) \sim D$ in (1.1) gives $\text{reg}(D, R^{-1}) = 0$. $\square$

We now prove the main bound (1.3) for an arbitrary regressor B . We first present two simple facts.

Lemma 1.6 (Per-point error). *Let*

$$\Delta_x(w) := \widehat{F}_x(w) - F_x(w).$$

Then for every w ,

$$|\Delta_x(w)|^2 = 4(B(x, w, \text{Re}) - \mu(x, w, \text{Re}))^2 + 4(B(x, w, \text{Im}) - \mu(x, w, \text{Im}))^2, \quad (1.10)$$

Moreover, one has

$$\mathbb{E}_w |\Delta_x(w)|^2 = 8 \text{reg}_x(D', B),$$

where reg_x is the binary regret conditioned on the context x .

Proof. By (1.8) and Lemma 1.4,

$$\text{Re } \Delta_x(w) = 2(B(x, w, \text{Re}) - 1/2) - 2(\mu(x, w, \text{Re}) - 1/2) = 2(B(x, w, \text{Re}) - \mu(x, w, \text{Re}))$$

and likewise

$$\text{Im } \Delta_x(w) = 2(B(x, w, \text{Im}) - \mu(x, w, \text{Im})).$$

Summing the squares of the real and imaginary parts gives the stated identity for $|\Delta_x(w)|^2$. Fix a context x , one has

$$\begin{aligned} \text{reg}_x(D', B) &= \frac{1}{2} \mathbb{E}_w [(B(x, w, \text{Re}) - \mu(x, w, \text{Re}))^2] + \frac{1}{2} \mathbb{E}_w [(B(x, w, \text{Im}) - \mu(x, w, \text{Im}))^2] \\ &= \frac{1}{8} \mathbb{E}_w |\Delta_x(w)|^2 \end{aligned}$$

The first step holds since $w \sim \text{Unif}(\mathbb{Z}_m^d)$, Re, Im is chosen uniformly at random, and the second step holds since Eq. (1.10). $\square$

Lemma 1.7. *For each fixed $\lambda \in \{0, \dots, m-1\}$, if $W \sim \text{Unif}(\mathbb{Z}_m^d)$ then $W + \lambda v_y$ is distributed as $\text{Unif}(\mathbb{Z}_m^d)$.*

Proof. For fixed λ and v_y , the map $W \mapsto W + \lambda v_y$ is a translation of the finite abelian group $\mathbb{Z}_m^d$, hence a bijection. $\square$

Lemma 1.8 (Regret transfer). *For every data distribution D and every binary regressor B , $\text{reg}(D, R^{-1}) \leq 8 \text{reg}(D', B)$.*

Proof. Fix x and a label y . Write the decode error of (1.9) as

$$E_{x,y}(W) := \text{Re} \left(\omega^{-\langle u_y, W \rangle} \frac{1}{m} \sum_{\lambda=0}^{m-1} \widehat{F}_x(W + \lambda v_y) \right) - P(y \mid x).$$

By Lemma 1.5, the F_x -part of $\widehat{F}_x$ reproduces $P(y \mid x)$ exactly, so only the error $\Delta_x = \widehat{F}_x - F_x$ remains:

$$E_{x,y}(W) = \text{Re} \left(\omega^{-\langle u_y, W \rangle} \frac{1}{m} \sum_{\lambda=0}^{m-1} \Delta_x(W + \lambda v_y) \right).$$

Using $|\text{Re } z| \leq |z|$, then $|\omega^{-\langle u_y, W \rangle}| = 1$, then Cauchy–Schwarz:

$$E_{x,y}(W)^2 \leq \left| \frac{1}{m} \sum_{\lambda=0}^{m-1} \Delta_x(W + \lambda v_y) \right|^2 \leq \frac{1}{m} \sum_{\lambda=0}^{m-1} |\Delta_x(W + \lambda v_y)|^2. \quad (1.11)$$

Take the expectation over $W \sim \text{Unif}(\mathbb{Z}_m^d)$. By Lemma 1.7 each summand has $W + \lambda v_y \sim \text{Unif}(\mathbb{Z}_m^d)$, so $\mathbb{E}_W |\Delta_x(W + \lambda v_y)|^2 = \mathbb{E}_w |\Delta_x(w)|^2$ for every λ ; averaging over λ and applying Lemma 1.6,

$$\mathbb{E}_W (\widehat{p}_y(x) - P(y \mid x))^2 = \mathbb{E}_W E_{x,y}(W)^2 \leq \mathbb{E}_w |\Delta_x(w)|^2 = 8 \text{reg}_x(D', B). \quad (1.12)$$

Finally, take the expectation over $(x, y) \sim D$, with the decoder's internal randomness W independent of (x, y) . The left side is $\text{reg}(D, R^{-1})$ by (1.1), and on the right $\mathbb{E}_{(x,y) \sim D} [\text{reg}_x(D', B)] = \text{reg}(D', B)$. Hence $\text{reg}(D, R^{-1}) \leq 8 \text{reg}(D', B)$. $\square$

Finally, we note R generates only one random data point, R^{-1} requires $2m = O(\log k)$ oracle calls, combining with lemma 1.8, this completes the proof of Theorem 1.2.

References

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